

HealthConnect Advanced Analytics & Decision Support Report

Week 6 — Data Analytics Track

Prepared by: Gladys Ogbeifun

AnalystLab Africa Data Analytics Programme

1. Executive Summary

This report presents the Week 6 advanced analytics work carried out on the HealthConnect appointment dataset (5,000 records). Building on the Week 5 baseline analysis, Week 6 deepened the investigation into appointment no-show behaviour through cross-segment analysis, KPI validation, and preparation of evidence for the Data Science track.

The overall observed no-show rate across the dataset is 48.46%, meaning almost half of all scheduled appointments were not attended. The Week 6 analysis confirms that this rate is not uniform across the patient base: it varies substantially according to booking lead time, previous no-show history, reminder status, distance from the facility, and appointment type.

The most significant finding is the strong interaction between previous no-show history and booking lead time. Appointments involving patients with two or more previous no-shows and a booking lead time of 31 or more days recorded a no-show rate of 72.96% — the highest rate identified in the analysis. This combination represents the clearest candidate for a high-risk appointment segment and is being handed off to the Data Science track for predictive validation.

All findings in this report are descriptive and diagnostic in nature. They identify meaningful associations in the data but do not establish causation. The Data Science track is expected to determine which of these relationships remain statistically significant and predictive once other variables are controlled for.

2. Week 5 → Week 6 Transition

Week 5 focused on preparing and analysing the HealthConnect appointment dataset, validating key KPIs, conducting exploratory analysis, and developing the initial Power BI dashboard. This work established a baseline understanding of appointment outcomes and identified important patterns associated with patient no-show behaviour.

The main Week 5 output was an analysed HealthConnect dataset supported by SQL analysis and Power BI visualisation. One of the most important findings was the high overall no-show rate of 48.46%, with booking lead time emerging as a particularly important pattern.

The main limitation identified from the Week 5 work is that descriptive analysis can identify patterns and relationships but cannot determine whether those relationships remain statistically meaningful when other factors are considered. It also cannot establish whether the identified variables can reliably predict future no-shows.

The Data Analytics and Data Science tracks are therefore directly relevant to the Week 6 activities. The Data Analytics track deepened and validated the findings from Week 5, while the Data Science track will investigate whether the identified factors have predictive value.

During Week 6, the focus was on advancing the existing analysis rather than repeating the Week 5 exploratory data analysis. The work included deeper cross-segment analysis, KPI validation, identification of high-risk appointment patterns, preparation of evidence for the Data Science track, refinement of business recommendations, and improvement of the existing dashboard based on the new findings.

3. Week 6 Integration Readiness

This section documents how the Week 6 Data Analytics output is structured to feed directly into the Data Science track.

Item	Detail
Integration track	Data Science (predictive modelling)

Item	Detail
Required input	Validated appointment dataset (5,000 records); cleaned and cross-tabulated variables: booking lead time, previous no-show history, reminder status, distance, appointment type
Analytical output provided	Segment-level no-show rates, interaction analysis, and a documented high-risk appointment profile
Integration objective	Enable the Data Science track to test whether the observed associations remain statistically significant and can support a predictive no-show risk model
Evidence of integration	This document, Section 10 (Data Science Handoff), together with the underlying validated dataset and Power BI dashboard

4. Data Validation

SQL validation was performed on the dataset prior to analysis to confirm completeness of the key analytical variables.

Validation Item	Result
Total records	5,000
Missing appointment outcome	0
Missing booking lead days	0
Missing booking lead-time group	0
Missing reminder value	0
Missing appointment type	0
Missing previous no-show history	0
Missing distance	90

The major analysis variables contain no missing values except distance, where 90 records have missing information. Distance-related findings are interpreted with this limitation in mind throughout the report.

5. Advanced Analytics

5.1 Booking Lead Time Analysis

Booking lead time demonstrated a strong observed relationship with appointment no-shows.

Booking Lead Time	No-Show Rate
0–7 Days	27.81%
8–14 Days	33.55%
15–30 Days	43.21%
31+ Days	60.49%

The difference between appointments booked 0–7 days in advance and those booked 31+ days in advance is 32.68 percentage points. This relationship warrants further investigation to determine whether booking lead time remains an important predictor after controlling for other variables.

5.2 Previous No-Show History

Previous no-show behaviour also demonstrated an observed relationship with future appointment attendance.

Previous No-Show History	No-Show Rate
No Previous No-Shows	43.51%
1 Previous No-Show	53.49%
2+ Previous No-Shows	61.02%

Patients with previous no-show history recorded higher observed no-show rates than patients without previous no-shows, supporting its inclusion in further predictive investigation.

5.3 Previous No-Show History × Booking Lead Time

The interaction between previous no-show history and booking lead time produced some of the highest observed no-show rates in the dataset.

Segment	No-Show Rate
2+ previous no-shows + 31+ days lead time	72.96%
1 previous no-show + 31+ days lead time	66.37%
No previous no-shows + 31+ days lead time	55.18%
2+ previous no-shows + 15–30 days lead time	57.62%
2+ previous no-shows + 0–7 days lead time	43.59%

This suggests that the interaction between previous attendance behaviour and booking lead time may be useful for further modelling investigation.

5.4 Reminder Analysis

Reminder Sent	No-Show Rate
No	51.39%
Yes	47.36%

Appointments with a recorded reminder had a 4.03 percentage-point lower observed no-show rate. This result should not be interpreted as proof that reminders caused the reduction; further statistical analysis is required to determine whether reminder status remains significant after controlling for other factors.

5.5 Distance Analysis

Distance Group	No-Show Rate
0–5 km	46.45%
6–10 km	47.01%
11–20 km	49.30%

Distance Group	No-Show Rate
21+ km	59.63%
Unknown	47.77%

The 21+ km group recorded the highest overall no-show rate at 59.63%. When combined with a booking lead time of 31+ days, the 21+ km group recorded a no-show rate of 71.15%. Because 90 distance values are missing, appropriate treatment of missing distance values should be considered during modelling.

5.6 Appointment Type × Booking Lead Time

Longer booking lead times were associated with higher no-show rates across all appointment types. For appointments booked 31+ days in advance:

Appointment Type	No-Show Rate
Follow-up	65.14%
Diagnostic Test	59.55%
Specialist Consultation	59.18%
General Consultation	58.10%

Follow-up appointments booked 31+ days in advance recorded the highest rate in this comparison at 65.14%.

6. Validated Findings

The following findings remained consistent and important after the deeper Week 6 cross-segment analysis:

- Booking lead time is strongly and consistently associated with no-show rate, rising from 27.81% (0–7 days) to 60.49% (31+ days).
- Previous no-show history is independently associated with future no-shows, rising from 43.51% to 61.02% as prior no-show count increases.
- The interaction of previous no-show history and booking lead time identifies the single highest-risk segment in the dataset (72.96%), exceeding either variable considered alone.
- Reminder status shows a modest but consistent association with lower no-show rates (a 4.03 percentage-point difference).
- Distance is associated with no-show rate, particularly beyond 21 km, and compounds with long booking lead times (71.15%).
- Appointment type interacts with booking lead time, with Follow-up appointments booked 31+ days out showing the highest rate among appointment types (65.14%).

These six relationships form the validated evidence base carried forward into the business recommendations and the Data Science handoff.

7. Refined KPIs

KPI	Value
Total appointments	5,000
No-show count	2,423
No-show rate	48.46%

KPI	Value
Attended count	2,314
Attended rate	46.28%
Cancelled count	263
Cancelled rate	5.26%

8. Business Impact Analysis

Among the validated findings, the following are assessed as having the greatest potential effect on HealthConnect operations:

- High-risk interaction segment (2+ previous no-shows + 31+ days lead time, 72.96%): the clearest and most actionable target for prioritised intervention, since it identifies a specific, sizeable segment rather than a broad population.
- Booking lead time overall: because it affects all appointment types and touches the majority of the schedule, adjustments to how far in advance appointments are booked could have the broadest operational reach.
- Distance combined with long lead time (71.15%): relevant to outreach and transport-related interventions for patients furthest from the facility.
- Follow-up appointments booked far in advance (65.14%): relevant to how follow-up care is scheduled and re-confirmed over time.

Reminder status, while statistically observable, currently shows the smallest effect size (4.03 percentage points) and is assessed as lower business impact until its independent effect is confirmed by the Data Science track.

9. Business Recommendations

9.1 Investigate Long Lead-Time Appointments

The analysis shows a substantial increase in observed no-show rates as booking lead time increases: 60.49% for appointments booked 31+ days in advance compared with 27.81% for appointments booked 0–7 days in advance. HealthConnect should investigate appointment-management strategies for appointments scheduled significantly in advance, particularly those exceeding 30 days.

9.2 Prioritise Patients With Previous No-Show History for Further Investigation

Patients with previous no-show history recorded higher observed no-show rates, rising from 43.51% (no previous no-shows) to 53.49% (one previous no-show) and 61.02% (two or more previous no-shows). This variable should be considered when developing future appointment-risk monitoring or predictive models.

9.3 Investigate High-Risk Appointment Combinations

The combination of previous no-show history and booking lead time produced particularly high observed no-show rates. Appointments involving patients with 2+ previous no-shows and a booking lead time of 31+ days recorded a 72.96% no-show rate. This group should be prioritised for further Data Science investigation to determine whether the combination can reliably identify higher-risk appointments.

9.4 Evaluate Reminder Strategies Further

Appointments with reminders recorded a lower observed no-show rate than appointments without reminders. However, the difference alone does not establish that reminders caused the reduction. HealthConnect should

evaluate reminder effectiveness further using statistical analysis and, where appropriate, controlled testing before making major operational changes.

9.5 Investigate Distance-Related Attendance Patterns

Appointments involving patients located 21+ km from the healthcare facility recorded a 59.63% observed no-show rate, increasing further when combined with a 31+ days booking lead time. HealthConnect should investigate whether distance represents a meaningful operational barrier and whether additional appointment-management strategies are appropriate for these patients.

9.6 Pay Attention to Long-Lead-Time Follow-Up Appointments

Follow-up appointments booked 31+ days in advance recorded the highest no-show rate among the appointment types analysed, at 65.14%. This pattern should be investigated further to determine whether follow-up appointments booked far in advance require different scheduling or follow-up strategies.

9.7 Use the Findings to Guide Predictive Modelling

The current analysis provides several candidate variables for Data Science investigation, particularly:

- Booking lead time
- Previous no-show history
- Reminder status
- Distance
- Appointment type
- Relevant interactions between these variables

The Data Science track should determine whether these variables provide statistically meaningful and predictive information before they are used for operational decision-making.

9.8 Avoid Treating Observed Associations as Causal Conclusions

The current findings should be used to guide further investigation rather than to make unsupported causal claims. The purpose of the Week 6 analysis is to identify meaningful patterns, validate the evidence, and provide a stronger foundation for predictive modelling and decision support.

10. Data Science Handoff

This section documents the findings provided to the Data Science track and the questions requiring further investigation, drawn from the Week 6 Data Science Handoff document.

10.1 Purpose of the Handoff

This handoff provides the Data Science track with the key findings from the Week 6 advanced analytics work on the HealthConnect appointment dataset. Its purpose is to identify relationships that require deeper statistical or predictive investigation and to provide evidence that can support modelling decisions. The findings represent observed patterns in the available dataset and should not be interpreted as proof of causation.

10.2 Questions for Data Science Investigation

1. Does booking lead time remain a significant predictor of no-shows after controlling for other available variables?
2. Does previous no-show history independently predict future no-shows?
3. Does the combination of previous no-show history and booking lead time provide additional predictive value?

4. Is reminder status statistically associated with appointment attendance after controlling for other relevant variables?
5. Does distance contribute meaningful predictive information?
6. Do appointment type, age, and other available patient or appointment characteristics improve no-show prediction?
7. Which variables contribute most strongly to the prediction of no-show behaviour?
8. Can the available variables support the development of a reliable and explainable no-show risk model?
9. Which modelling approach provides the most appropriate balance between predictive performance and interpretability?
10. How should the model be evaluated to ensure that it is appropriate for the HealthConnect use case?

10.3 Recommended Data Science Investigation

The Data Science track may consider:

- Reviewing the quality and consistency of the modelling variables.
- Defining the modelling target clearly.
- Assessing how cancelled appointments should be treated.
- Checking the class distribution.
- Investigating potential feature interactions.
- Establishing a baseline model.
- Testing appropriate classification models.
- Evaluating model performance using suitable classification metrics.
- Reviewing feature importance and model explainability.
- Assessing whether the model generalizes beyond the available sample.
- Identifying limitations before any operational use is recommended.

10.4 Expected Data Science Contribution

The Data Science track is expected to determine:

- Which analytical findings remain statistically meaningful.
- Which variables should be considered for modelling.
- Whether important interactions exist between variables.
- Whether the dataset can support reliable no-show prediction.
- Which model is most appropriate.
- How accurately the model performs.
- Which factors contribute most to predicted risk.
- Whether the model is sufficiently interpretable for business use.

10.5 Business Relevance

If the identified patterns remain meaningful after statistical and predictive validation, HealthConnect could potentially use the findings to identify appointments that may require additional attention. Possible decision-support applications could include prioritising appropriate appointment-management interventions for higher-risk appointments. Any operational recommendation should only be made after the Data Science investigation confirms that the identified relationships are sufficiently reliable.

11. Dashboard Improvements

Week 6 Dashboard Improvement

The Week 5 Power BI dashboard provided a descriptive view of HealthConnect appointment outcomes and no-show patterns across individual dimensions, including appointment type, appointment time, appointment day, age group, distance group, booking lead time, previous no-show history and reminder status.

During Week 6, the dashboard was improved to reflect the deeper analysis conducted through SQL. Rather than recreating the existing Week 5 visuals, additional cross-segment analysis was incorporated to examine how important variables interact.

The Week 6 dashboard improvements focused on:

- No-show rate by previous no-show history and booking lead time.
- No-show rate by booking lead time and reminder status.
- No-show rate by appointment type and booking lead time.

These additions provide a more detailed view of high-risk appointment patterns and support decision-making beyond the individual-factor analysis completed during Week 5.

The most significant finding from the advanced analysis was the combination of previous no-show history and booking lead time. Appointments involving patients with two or more previous no-shows and a booking lead time of 31 or more days recorded a 72.96% no-show rate.

The dashboard improvements therefore support the Week 6 objective of deepening and validating existing findings rather than repeating the original exploratory analysis.

12. Data Limitations

- Missing distance values: 90 of 5,000 records (1.8%) have missing distance information; distance-related findings, including the combined distance/lead-time result, should be interpreted with this limitation in mind.
- Descriptive nature of the analysis: the current analysis is primarily descriptive and diagnostic rather than inferential or predictive.
- No causal conclusions: none of the relationships identified — including reminder status and booking lead time — should be interpreted as proof of causation. Other factors may influence both the predictor and the outcome.
- Dataset limitations: the analysis is based on a single dataset of 5,000 records from HealthConnect; findings may not generalise to other periods, locations, or patient populations without further validation.
- Potential modelling limitations: class balance (48.46% no-show vs. 46.28% attended vs. 5.26% cancelled), treatment of cancelled appointments, and possible feature interactions will need to be assessed before a predictive model can be considered reliable.
- Other limitations: variables outside the current dataset (for example, more detailed patient demographics or clinical context) were not available for this analysis and may affect predictive performance.

13. Cross-Track Contribution

The Data Analytics work in Weeks 5 and 6 supports the Data Science track in the following specific ways:

- Provides a validated, cleaned dataset with confirmed completeness across all major analytical variables (Section 4).
- Identifies and quantifies the individual variables most strongly associated with no-shows, narrowing the candidate feature set for modelling.

- Surfaces a specific, high-value feature interaction (previous no-show history $\times$ booking lead time) that the Data Science track can test explicitly rather than discover from scratch.
- Frames ten specific investigation questions (Section 10.2) that give the Data Science track a clear, business-relevant starting point.
- Documents known data limitations (Section 12) up front, so modelling decisions — such as how to treat missing distance values or cancelled appointments — can be made deliberately rather than discovered mid-analysis.
- Provides a dashboard that can be extended to display model outputs (e.g., predicted risk scores) once the Data Science track produces them.

14. Conclusion

The Week 6 advanced analytics work confirms that HealthConnect's overall no-show rate of 48.46% is not evenly distributed across appointments. Booking lead time, previous no-show history, distance, reminder status, and appointment type are all associated with meaningfully different no-show rates, and several of these variables compound when combined.

The most important single finding is that appointments with patients who have two or more previous no-shows and a booking lead time of 31 or more days show a 72.96% no-show rate — nearly three times the rate of appointments booked within 7 days by patients with no prior no-show history (27.81%). This gives HealthConnect a concrete, evidence-based starting point for prioritising appointment-management interventions.

These findings are descriptive rather than causal and are now being handed off to the Data Science track for statistical and predictive validation, as documented in Section 10. Once validated, they can support the development of a no-show risk model and more targeted, evidence-based scheduling and outreach decisions.

15. Week 7 Analytical Testing Requirements

The following items are proposed for testing and validation in Week 7:

- Predictive model results: initial output from the classification model(s) tested by the Data Science track.
- Model performance: evaluation using suitable classification metrics (e.g., accuracy, precision, recall, F1, ROC-AUC) appropriate to the class distribution.
- Feature importance: confirmation of which variables (booking lead time, previous no-show history, distance, reminder status, appointment type, and their interactions) contribute most to predicted risk.
- Statistical significance where applicable: formal testing of the associations identified in Sections 5 and 6.
- Dashboard validation: confirmation that dashboard figures remain consistent with the validated dataset and any model outputs added to it.
- Business usability: assessment of whether the model's outputs are interpretable and actionable enough for operational use by HealthConnect staff.
- Consistency between analytical and modelling outputs: verification that the Data Science findings align with, or appropriately refine, the Week 6 descriptive findings in this report.